Henry Wang

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EDUCATION

Stanford University **Stanford, CA**

B.S. Math | Expected Graduation: June 2029 GPA: 3.92

- **Relevant Courses:** Machine Learning, Decision Making under Uncertainty, Probability for Computer Scientists, Game Theory, Data Structures and Algorithms, Efficient Generative Language Models

EXPERIENCE

Stanford Safe and Intelligent Autonomy Lab | Drone Avoidance Demo **June 2026 - Present**

Student Research Assistant

- Trained a vision-based autonomous drone controller, improved collision-free flight success from 16.7% to 83.3% across 30 simulated flight trials per evaluation
- Finetuned DINOv2 computer vision model to predict safety of drone actions using 10 rounds of DAgger (100M+ state-action datapoints), improving prediction R^2 from 0.44 to 0.945
- Identified controller intervention frequency as a performance bottleneck; increasing frequency (increased intervention threshold from 0.15 to 0.5) improved flight success from 66.7% to 83.3% without model retraining

Stanford Radio Club | Pepperwood Air Quality Dashboard **September 2025 - Present**

Data Analyst

- Analyzed environmental data from radio-connected sensors to identify air-quality patterns during prescribed burns; presented findings at AGU25 and accepted for presentation at AGU26
- Applied XGBoost feature importance and correlation analysis to identify redundant sensor measurements, reducing power consumption by 43% while preserving correlation with the original signal ($r > 0.7$)
- Built 10+ maps using QGIS, Folium; deployed 1 to air-quality dashboard, delivering insights to local firefighters

UNC Wearable Bioelectronics Lab | Blood Pressure Device Demo **June 2023 - April 2025**

Student Research Assistant

- Co-authored an IEEE paper on CNN-LSTM blood pressure estimation from photoplethysmography ($r > 0.7$)
- Built a Pandas data processing pipeline with bandpass and moving-average filtering, cleaning and transforming 200+ minutes of clinical trial photoplethysmography data into 1,200 model training-ready inputs.
- Addressed IEEE peer-review feedback by evaluating feature relevance using saliency tests ($p < 0.001$), Gaussian-noise testing (ICC > 0.7), and ridge regression against gold-standard Vicorder PPG ($R^2 = 0.70$)

PROJECTS

Tree-of-Thoughts for Math Theorem Proving via Lean | Python, Google AI Studio, Lean 4

- Built a Python math theorem-proving system combining Gemini 2.5 Flash, Tree-of-Thoughts search, and Lean 4 proof verification via Pantograph (Lean verifier in Python)
- Implemented beam search algorithm that revisited promising proof branches, improving benchmark solve rate from 8.7% to 37.3% (11 to 47 out of 126 total problems) with equal compute

TECHNICAL SKILLS

- **Languages:** Python, Java, C, MATLAB, SQL, R
- **Machine Learning Frameworks & Tools:** TensorFlow, PyTorch, scikit-learn, CUDA

LEADERSHIP & OTHER INTERESTS

Stanford Boxing Club | *Financial Officer* | 2026 Stanford Fight Night (my fight)

- Managed club finances and helped organize the inaugural 2026 Stanford Fight Night, which raised ~$5,000; currently planning the 2027 event and expanding participation to 5 additional colleges

Weightlifting | 4 years

- Tracked calories in bulk/cut (ate ~96 lbs of chicken over 9 months); 415 squat, 225 bench at 185lbs bodyweight